Impact of the transplantation on the mortality of kidney transplant candidates: a cohort-based study based on dynamic propensity scores matching

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Université de Poitiers   CHU Poitiers   Inserm La science pour la santé From science to health

Plan

Context

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Results

Discussion

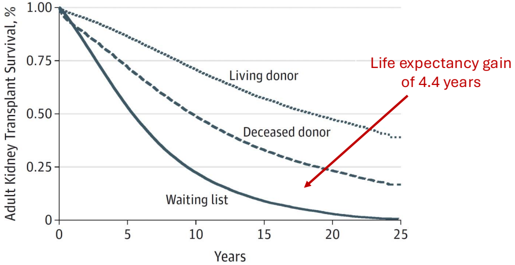
The benefit of kidney transplantation

- Two treatments of end-stage renal disease : long-term dialysis and kidney transplantation (KT).
- KT is considered to be the best treatment since studies reflected its benefit in terms of life expectancy.

JAMA Surgery ; 2015 ; 150(3) :252-259.1

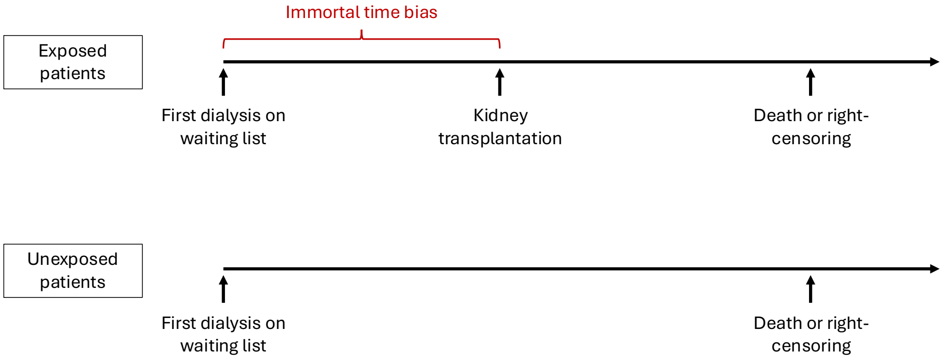
Survival Benefit of Solid-Organ Transplant in the United States

Abbas Rana, MD; Angelika Gruessner, PhD; Vatche G. Agopian, MD; Zain Khalpey, MD, PhD; Irbaz B. Riaz, MBBS; Bruce Kaplan, MD; Karim J. Halazun, MD; Ronald W. Busuttil, MD, PhD; Rainer W. G. Gruessner, MD



No. at risk						
Waiting list	239 162	43 202	7083	1472	275	32
Deceased donor	148 292	74 561	27 850	8576	1859	16
Living donor	87 387	50 125	19 255	5275	1115	8

An overestimation of the kidney transplantation benefit



Wolfe et al., New England Journal of Medicine, 1999.

- The use of an upated Cox model with the transplantation as a time-dependent exposure from the first dialysis on the waiting list.

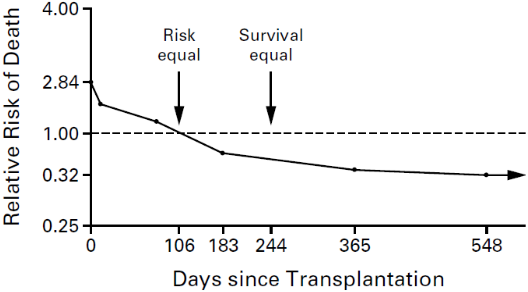
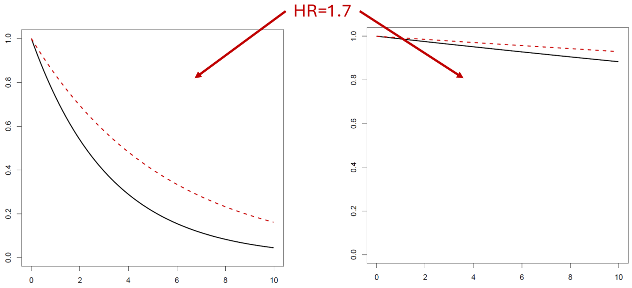


Figure 2. Adjusted Relative Risk of Death among 23,275 Recipients of a First Cadaveric Transplant.

The main limits of results from such an updated Cox model

- Post-registration confounders may bias the results :
  - Post-registration deterioration of health deacreases the chance of transplantation.
  - It may overestimate the transplantation benefit.
- One can expect an overestimation of the KT effect : patients with a deteriorating health are less likely to be transplanted.
- The magnitude of the transplantation effect is difficult to interpret from Hazard Ratio (HR).

Interpretation of Hazard Ratios



Objective of our study

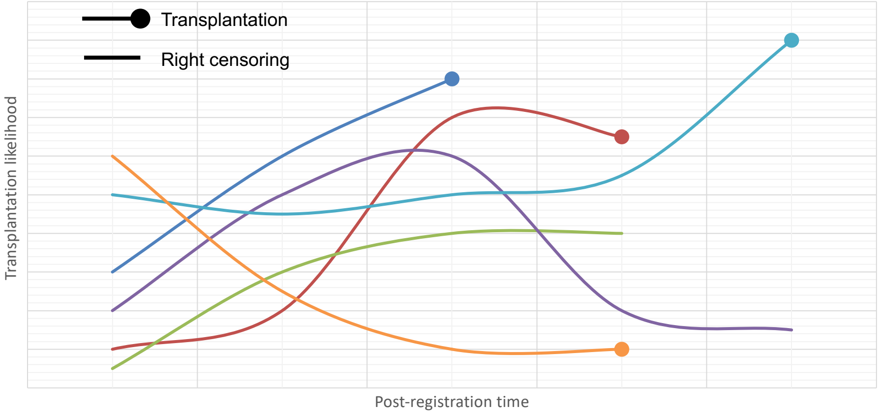
- To better appraise the life expectancy of KT recipients versus similar patients waiting for KT at the same time post-registration.

Plan

The study population (N=23,128)

- Patients from the French REIN registry.
- Patients registered for kidney transplantation only.
- Adults registered for the first time on a waiting list and with at least one dialysis session after registration (exclusion of pre-emptive transplantation).
- Patients with a first dialysis from 1 January 2005 to 31 December 2018
- Patients with a follow-up.

We adapted the method proposed by Lu (Biometrics ; 2005)



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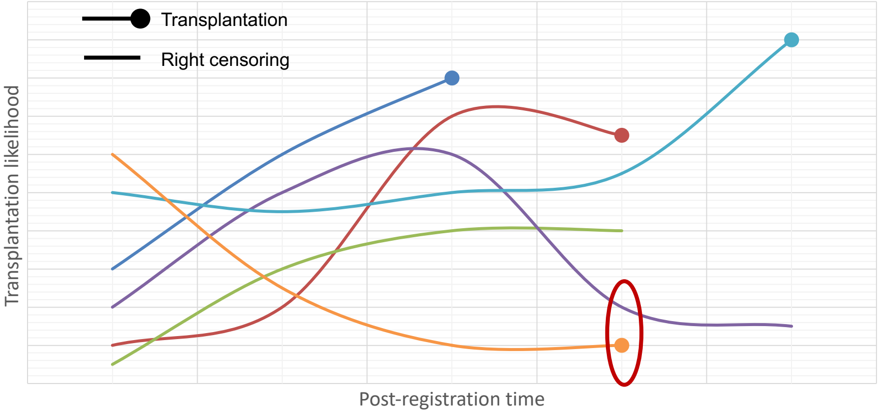

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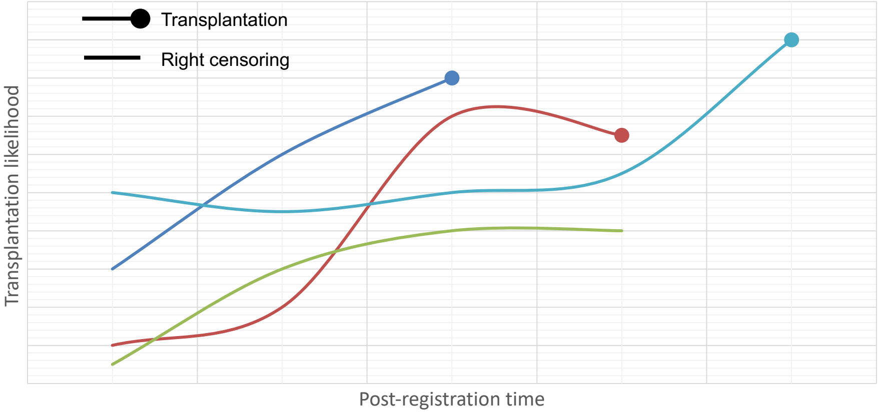

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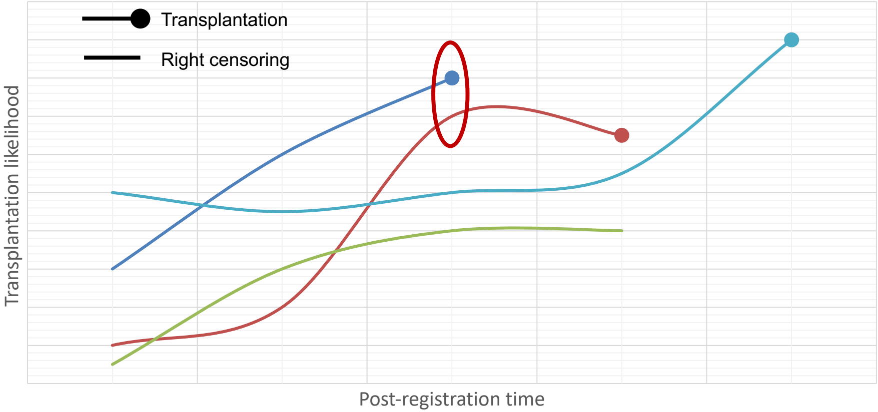

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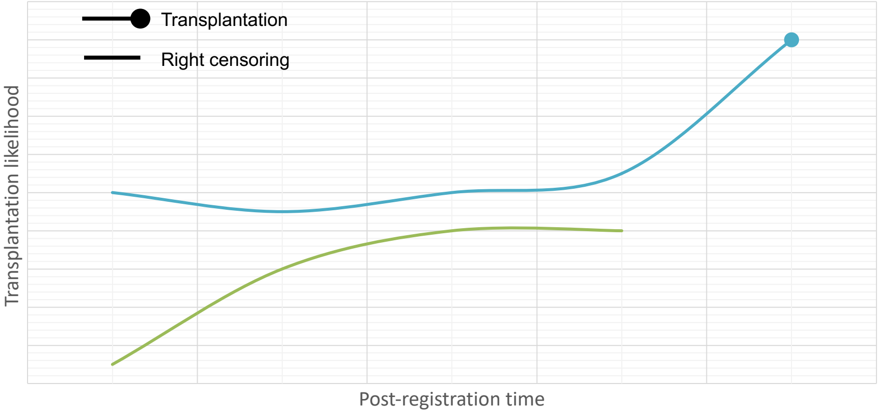

The first step : estimating the time-dependent likelihood of transplantation

- The hazard of receiving a kidney transplant at any times was estimated by using an updated Cox model.
- Death and waiting list removal were right-censored.
- We considered the following 10 covariates :
  - Time-fixed variables : age, sex, blood group, time from dialysis to registration, year of registration on the waiting list, duration of temporary contraindication, history of diabetes and living area.
  - Time-dependent variables : Body mass index, calculated panel reactive antibody.

The second step : matching pairs

- We used the nearest neighbor matching algorithm with a maximum caliper of 0.20.
- Exact matching was considered for the unbalanced covariates.
- We used random matching without replacement, meaning that once matched, a patient could not be matched again in another pair.

The third step : analysing the matched cohort

- The matching time corresponded to the pseudo-randomization between transplantation and waiting.
- From this baseline, the outcome was the time-to-death.
- We obtained the survival curves by using the Kaplan-Meier estimator.
- The mean estimations and the related confidence intervals were derived by bootstrapping the full 3-step procedure.
- We respected an intention-to-treat approach : transplanted patients inthe waiting group were not censored/excluded.
- The cumulative probability of KT in the awaiting group was estimated by the Aalen-Johansen method.

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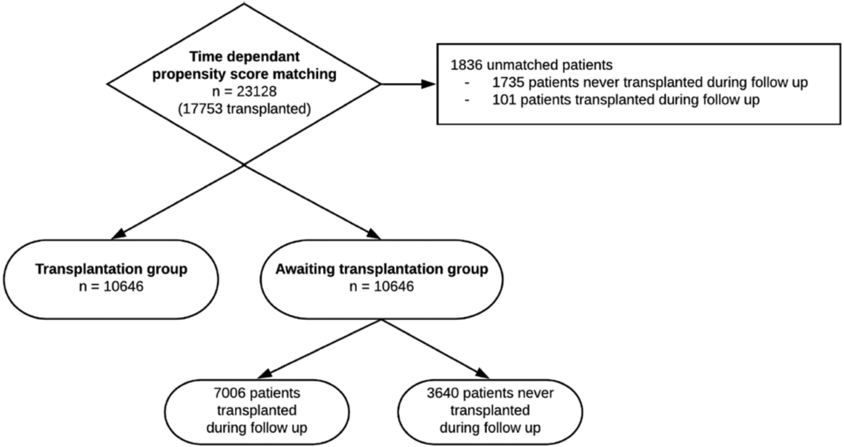
Flowchart


Time dependant
propensity score matching
n = 23128
(17753 transplanted)
1836 unmatched patients
- 1735 patients never transplanted during follow up
- 101 patients transplanted during follow up
Transplantation group
n = 10646
Awaiting transplantation group
n = 10646
7006 patients
transplanted
during follow up
3640 patients never
transplanted
during follow up

Description of the unmatched patients

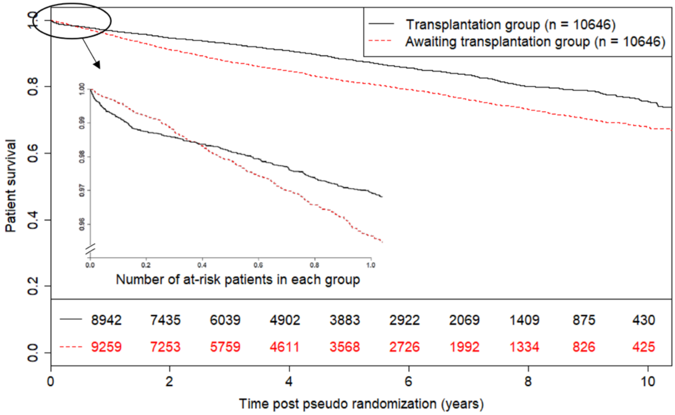
Median [interquartile range] Effective (%)	Unmatched patients (N=1836)	Matched patients (N=21292)	p-value
Recipient age (years)	57.7 [49.3;64.2]	54.8 [43.9;63.4]	<0.01
Male recipient	1184 (65%)	13635 (64%)	0.72
Recipient BMI (kg/m²)	26.0 [22.8;30.1]	24.9 [22.1;28.4]	<0.01
Year of inscription on waiting list			<0.01
- 2005-2007	48 (2.6%)	2702 (13%)	
- 2008-2010	179 (9.7%)	5830 (27%)	
- 2011-2013	448 (24%)	7016 (33%)	
- 2014-2016	1161 (63%)	5744 (27%)	
Primary cause of end stage renal disease			<0.01
- Diabetic nephropathy	446 (24%)	2605 (12%)	
- Hypertension	351 (19%)	3237 (15%)	
- Other	238 (13%)	3400 (16%)	
- Polycystic Kidney Disease	198 (11%)	3480 (16%)	
- Primary glomerulopathy	283 (15%)	5034 (24%)	
- Pyelonephritis	78 (4.2%)	1037 (4.9%)	
- Unknown	242 (13%)	2499 (12%)	
Recipient blood group			<0.01
- A	574 (31%)	8428 (40%)	
- AB	61 (3.3%)	874 (4.1%)	
- B	314 (17%)	2546 (12%)	
- O	887 (48%)	9444 (44%)	
Dialysis technique			<0.01
- Hemodialysis	1630 (89%)	17981 (85%)	
- Peritoneal dialysis	206 (11%)	3310 (16%)	
Time from dialysis to inscription (days)	245.0 [21.5;496.0]	193.0 [-1.0;414.0]	<0.01
Calculated Panel Reactive Antibody > 0%	1270 (69%)	7835 (37%)	<0.01
Potential matched donors	277.0 [93.0;518.2]	424.0 [179.0;634.0]	<0.01
Positive anti-HLA class I antibodies	513 (28%)	4453 (21%)	<0.01
Positive anti-HLA class II antibodies	315 (17%)	2923 (14%)	<0.01
History of diabetes	724 (39%)	4412 (21%)	<0.01

Well-balanced characteristics at the matching time

TABLE 2. Patient Characteristics in the Transplantation and Awaiting Transplantation Groups at the Time of Pseudo-Randomization

	Awaiting Transplantation Group (N = 10,646)	Transplantation Group (N = 10,646)	Standardized Difference (%)
Recipient age (years), median (IQR)	56.9 (45.8 to 65.1)	55.3 (44.5 to 64.0)	8.6
Male recipient, effective (%)	6782 (64)	6853 (64)	1
Recipient BMI (kg/m^2), median (IQR)	25.1 (22.3 to 28.6)	25.0 (22.1 to 28.4)	3.7
Year of inscription on waiting list, effective (%)			0
2005–2007	1351 (13)	1351 (13)	
2008–2010	2915 (27)	2915 (27)	
2011–2013	3508 (33)	3508 (33)	
2014–2016	2872 (27)	2872 (27)	
Primary cause of end-stage renal disease, effective (%)			4.3
Diabetic nephropathy	1316 (12)	1289 (12)	
Hypertension	1688 (16)	1549 (15)	
Other	1688 (16)	1712 (16)	
Polycystic kidney disease	1727 (16)	1753 (17)	
Primary glomerulopathy	2453 (23)	2581 (24)	
Pyelonephritis	517 (4.9)	520 (4.9)	
Unknown	1257 (12)	1242 (12)	
Recipient blood group, effective (%)			42
A	3243 (30.5)	5185 (48.7)	
AB	327 (3.1)	547 (5.1)	
B	1477 (13.9)	1069 (10.0)	
O	5599 (52.6)	3845 (36.1)	
Dialysis technique, effective (%)			3.1
Hemodialysis	9053 (85)	8928 (84)	
Peritoneal dialysis	1592 (15)	1718 (16)	
Time from dialysis to inscription (days), median (IQR)	201.0 (2.0 to 424.0)	185.0 (−3.0 to 404.0)	2.2

Survival curves and *HR*



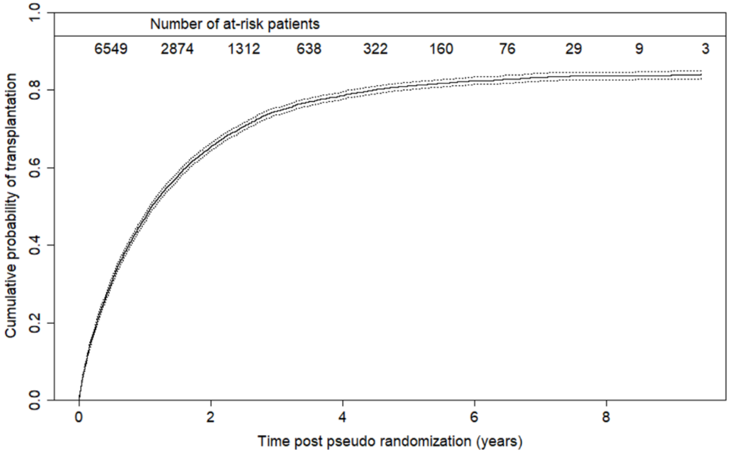
- The HR was 1.45 (95% CI = 1.11-1.90) during the first 3 months.
- The HR was 0.6 (95% CI = 0.55-0.65) afterwards.

Mean life expectancy according to several follow-up times

	Transplantation Group Mean Life Expectancy		Awaiting Transplantation Group Mean Life Expectancy		Mean Life Expectancy Difference	
Time (years)	**RMST**	**95% CI**	**RMST**	**95% CI**	**ΔRMST**	**95% CI**
1	0.98 years	0.98, 0.98	0.98 years	0.98, 0.98	0.03 months	–0.01, 0.08
3	2.9 years	2.9, 2.9	2.8 years	2.8, 2.8	0.93 months	0.7, 1.1
5	4.7 years	4.7, 4.7	4.5 years	4.5, 4.5	2.4 months	2.0, 2.9
10	8.8 years	8.7, 8.9	8.2 years	8.1, 8.3	6.8 months	5.5, 8.2

RMST indicates restricted mean survival time.

Probability to be transplanted in the awaiting transplantation group



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Interpretation of the results

- The current litterature :
  - The HR varies according to the time post-transplantation.
  - It ranges from 0.25 to 0.32 beyond 1-year post-transplantation.
  - It does not reflect the magnitude of the kidney transplantation effect.
- Our proposal :
  - The life expectancy gain due early KT was 6.8 months (95% CI = 5.5-8.2).
  - The proportion of transplanattion was higher than 60% in the awaiting transplantation group at 2 years.
  - It confirms the benefit of early KT when possible.

Main limitations of the results

- We did not compared KT versus long-term dialysis but early versus delayed KT.
- We did not consider the quality of life.
- We may have missed possible confounders.

References

- Rana et al. Survival benefit of solid-organ transplant in the United States. JAMA Surg. 201 ;150(3) :252-9.
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- Lu. Propensity score matching with time-dependent covariates. Biometrics. 2005 ;61 :721-728.
- Lenain et al. Clinical Trial Emulation by Matching Time-dependent Propensity Scores : The Example of Estimating Impact of Kidney Transplantation. Epidemiology. 2021 ;32(2) :220-229.

Slides available at https ://github.com/chupverse/slides-of-seminars